S12. Capacity of a specified finite noisy genotype–phenotype channel

Pure Thought research portfolio

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Source: synthesis challenge 30; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

Suppose the same genetic sequence can produce different modeled outcomes because of noise. How reliably can an observer infer the chosen sequence from the outcome? This problem treats the relationship as an explicitly given communication channel and asks for certified bounds on the greatest information it can transmit under any stated input-cost limit.

Key terms

- **Genotype** — An input genetic sequence or label in the finite model.
- **Phenotype** — An output trait or state as defined by the model.
- **Channel law** — The conditional probability of each output given each input.
- **Noise** — Variation in the output for a fixed input, as encoded by that channel law.
- **Input distribution** — The probabilities assigned to selecting the possible genotypes.
- **Mutual information** — A measure of how much observing the output reduces uncertainty about the input.
- **Shannon capacity** — The largest mutual information over the allowed input distributions for this channel.
- **Bit** — The information unit obtained when logarithms use base two.
- **Cost constraint** — A bound on average input cost, such as a prescribed sequence-dependent resource measure.
- **Relative entropy** — A measure of discrepancy between probability distributions, also called Kullback–Leibler divergence.

Why this matters

An explicit channel separates the number of possible outputs from their reliability and distinguishability. Capacity bounds give a rigorous way to compare modeled noise mechanisms or resource constraints, and to check whether a numerical optimization has found the best input distribution.

What progress would mean

Progress would bracket capacity with a feasible input distribution and a valid upper certificate. For a deterministic unconstrained channel, the number of reachable outputs supplies a simple benchmark. The answer describes the specified probabilities and costs; it is not an intrinsic information value of a genome or a measure of evolutionary fitness.

Revised core question

For a finite conditional phenotype law P(y|g) and optional genotype cost constraint, what certified lower and upper bounds can be placed on Shannon capacity?

Precise scope, target and outputs

Supply finite genotype and phenotype alphabets, a stochastic matrix W(y|g), and optional costs c(g) with E[c]≤C. Define capacity max_p I_p(G;Y), with logarithms base two. A chosen input distribution gives a lower bound. For the unconstrained channel, any output distribution q with adequate support gives upper bound max_g D(W(.|g)||q); constrained dual bounds must include the cost multiplier. Use interval logarithms and validated probabilities. Connect to PRD 30 by constructing W from a fully declared folding/noise rule.

Detailed critique of the synthesis framing

Mutual information is a property of a joint distribution, not a map alone. The original omits the input distribution, noise, phenotype alphabet and optimization constraints. For a deterministic finite map with unrestricted input distribution, capacity is simply $\log_2$ of the number of reachable phenotypes, so a purported hard capacity problem may collapse. Fano/Le Cam bounds answer specific inference questions and are not automatic capacity certificates. The rewrite specifies a nontrivial noisy channel, separates achievability from converse bounds, and keeps biophysical interpretation conditional on the chosen channel.

Validation and milestones

Baseline: binary symmetric-channel capacity and deterministic-map capacity reproduced. Strong: a certified bracket for a finite GP channel and a cost-constrained extension with a valid dual. Check rows sum to one, handle zero probabilities explicitly, and report unresolved numerical gaps.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$C = \max_p I_p(G;Y), \qquad I_p(G;Y) \le C \le \max_g D(W(\cdot|g)\|q).$$

For a cost limit c_*, any $\lambda \ge 0$ and adequately supported q give

$$C(c_*) \le \lambda c_* + \max_g \{D(W(\cdot|g)\|q) - \lambda c(g)\}.$$

Use base-two logarithms and verified stochastic rows. A feasible input distribution supplies a lower bound, and interval logarithms certify numerical values. For a deterministic unrestricted finite map, capacity is $\log_2 |\operatorname{im} \Phi|$; this differs from neutral-network connectivity.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise “publication-quality” in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.